Chapter 1:

INTRODUCTION OF INDUSTRY:

Compiler's Technology, as the name suggests, is intricately tied to the world of programming. It's the art and science of crafting software that takes code written by humans in high-level programming languages and transforms it into instructions a computer can execute. In essence, it's the bridge between human creativity and the digital world.

The proposed System is the e-commerce website HTML, CSS, JavaScript and Java programming language (servlet and jsp) is used to develop this project. Compilers Technologies provide industry-oriented training which helps to integrate academics with real time corporate world. Their Mission is to develop the technically sound, well skill students for industries and create maximum employment. Compiler Technology is grooming software product Development Company. They develop software products which help society to make life easy and interesting. Their team is always dedicated for the Development and Innovations. Their research in a product development makes the product outlined as per client requirement. Their development strategy gives throughput which inclined with the user requirements. Their product development starts from user's requirement and ends on efficient and well develop product. Their team has well experience and skilled experts who are able to give efficient and in time solution. Every solution provided by them is unique and innovative. Their approach has been always customer and market communication centric. They create products keeping our target consumer and audience in mind. They expertise always focuses on market requirement and development.

• *What is Project Development Training Program?*

They have started project development training program for the Engineering, Polytechnic as well as for Professional students. The concept behind this program is to provide Practical project development training, so that students get interact with latest project development strategy and emerging technologies. Their vision behind this program is to build the students technical skill required for the actual project development in industry.

Why Compiler's Technology Matters

The significance of Compiler's Technology lies in its role as an enabler. It's the technology that empowers developers to write code in familiar, expressive languages like Python, Java, or C++, and then translates it into efficient machine code. Without it, the software we use every day, from apps on our smartphones to the software running in our cars, would be a mirage.

The Student Advantage

What makes this internship even more exciting is its focus on students. It's designed for those who want to expand their horizons and are eager to dive into the world of programming, compilers, and software development. Here, you'll be mentored and guided by professionals who understand the unique challenges and opportunities that students bring.

What Awaits You

During your internship in Compiler's Technology, you'll have the chance to explore various aspects of programming and software development. Whether you're interested in optimizing compilers for new programming languages, diving into code security, or exploring the latest trends in Just-In-Time (JIT) compilation, there's a wide array of opportunities for you to discover and grow.
Throughout this educational journey, students will have the chance to explore a plethora of opportunities within Compilers Technology. They will dive into the intricacies of optimizing compilers for modern programming languages, investigate techniques to enhance code security, and embark on a voyage into the realm of Just-In-Time (JIT) compilation for dynamic languages. The educational experience will be a diverse and enriching one.

Shaping the Future IT Workforce

Our mission extends beyond theoretical knowledge. We aspire to nurture a generation of students who not only understand Compilers Technology but also actively contribute to its growth. Students will not only acquire skills in designing compilers but will also gain insights into their practical applications across various industries. Our goal is to prepare students to be future leaders and innovators in IT.

In conclusion, your journey into the world of Compiler's Technology within the low-scale industry is a unique and promising one. It's an opportunity to learn, grow, and shape your future in a field that powers the digital world. As you embark on this internship, remember that in the world of compilers and programming, small-scale industries hold immense potential, and your contributions can make a significant impact. Welcome to the fascinating world of Compiler's Technology, where code becomes reality, and your journey begins.

Chapter 2:

SOFTWARE AND REQUIREMENTS:

Scope of Work

The proposed System is the **BLOOD BANK SYSTEM** work is in development stage. As this project will be launched as product and it is a big software so it will take surely some more months to complete. The expert developers from Compilers Technologies are working on this project and by applying their full efforts trying to end the project as soon as possible. When it comes to technologies obviously many advance web technologies are going to be used while developing this project.

Operating Environment

Software required while development:

- OS: Windows 11
- Language: Advance Java
- Database: MySQL
- Libraries: Mysql-connector-java-8.0.30.jar, JCalender-1.3.2.jar
- IDE: Apache NetBeans IDE 13, NotePad++

Hardware required while development:

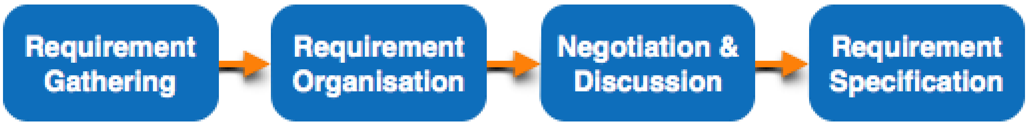
- Minimum RAM:-512 MB upto Maximum RAM:-8 GB
- Minimum Hard Disk:-10 GB upto Maximum Hard Disk:-80 GB
- Processor:-Intel Pentium 4 upto Maximum Processor:- Core i7
- Minimum GPU:- 1 GB upto Maximum GPU 8 GB
- Monitor: - 15/17 inch (color)

- Keyboard: - Standard

User Requirements

Requirements only tell you what your application must have and what it must allow users to do. Requirements do not tell you how to design or develop the site to have those features, functions, and content. The other design steps help you figure out how to make sure that the site is organized, written, and designed to satisfy the requirements.

Software Requirements



Requirements gathering - The developers discuss with the client and end users and know their expectations from the software.
Organizing Requirements - The developers prioritize and arrange the requirements in order of importance, urgency and convenience.
Negotiation & discussion - If requirements are ambiguous or there are some conflicts in requirements of various stakeholders, if they are, it is then negotiated and discussed with stakeholders. Requirements may then be prioritized and reasonably compromised.
The requirements come from various stakeholders. To remove the ambiguity and conflicts, they are discussed for clarity and correctness. Unrealistic requirements are compromised reasonably.
Requirement Specification - All formal & informal, functional and non-functional requirements are documented and made available for next phase processing.

Software Requirements

NetBeans:

NetBeans is an integrated development environment (IDE) for Java. NetBeans allows applications to be developed from a set of modular software components called modules. NetBeans runs on Windows, macOS, Linux and Solaris.

JDK 19:

JDK 19 delivers language Improvements from OpenJDK project Amber (Record Patterns and Pattern Matching for Switch); library enhancements to interoperate with non-Java Code (Foreign Function and Memory API) and to leverage vector instructions (Vector API) from OpenJDK project Panama; and the first previews for Project Loom (Virtual Threads and Structured Concurrency), which will drastically reduce the effort required to write and maintain high-throughput, concurrent applications in Java.

MYSQL:

MySQL is one of the most recognizable technologies in the modern big data ecosystem. Often called the most popular database and currently enjoying widespread, effective use regardless of industry, it’s clear that anyone involved with enterprise data or general IT should at least aim for a basic familiarity of MySQL. With MySQL, even those new to relational systems can immediately build fast, powerful, and secure data storage systems. MySQL’s programmatic syntax and interfaces are also perfect gateways into the wide world of other popular query languages and structured data stores.

MYSQL command line client/MYSQL workbench:

MYSQL command line client

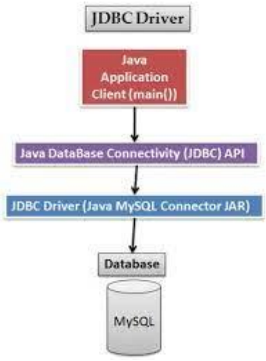
The MySQL is a simple SQL shell that has input line editing capabilities. It supports interactive and no interactive usage. When it is used interactively, query results are presented in an ASCII-table format.

MYSQL workbench

MySQL Workbench is a unified visual database designing or graphical user interface tool used for working with database architects, developers, and Database Administrators. It is developed and maintained by Oracle.

MYSQL Connector:

MySQL provides connectivity for Java client applications with MySQL Connector/J, a driver that implements the Java Database Connectivity (JDBC) API. The API is the industry standard for database-independent connectivity between the Java programming language and a wide range of – SQL databases, spreadsheets etc.



Chapter 3:

Activities and Processes:

First here we discussed the topic which we are going to select and as soon as selection we discussed the metadata required for our project and decides the things which will done by our software.

Once all this type of discussion was done, we. Then we divide our project in small modules and distributed between each member and once whole module is created, we have joint them properly and performed our whole testing by using the concept of white and black box testing.

1. **Login Page:**

   First of all, we have created a login page which will appear when you will open the application. This page is designed to save the data present in a software from an unauthorized person. In this page we have two fields username and password. When we enter incorrect username and password it will show a popup message incorrect username or password. If we enter correct username or password we will move to next page. It having one more button for exit this application.

2. **Home Page:**

   After login this one is first page we will see. In this page we have four button and each button having different functionality (i.e Donar, Receiver, Importance of blood, view Tables). From a home page user can go anywhere he wants to go which makes an user to operate our software easily and more efficiently. This home was designed in such a way which can be used by anyone.

3. **Blood Donation Check Page:**

This is a page which will open when you click on donor page. Before adding donor in our database, it is must important to fill information in this page because This page is specially designed to be secured from receiving blood from unhealthy person before entering donation page, we have to fill all the attributes of your blood if they all are proper then you will redirect to donor page. If any attribute among the all is not proper then I will highlight that with red color.

4. **Donar Page:**

Once you successfully filled the blood donation check page then you will be redirected to Donor page. This a page in which we can add the details of a donor. In details here we need to take id name and blood group as an input and it is mandatory to give the amount of blood that is going to donated by donor as later confirmation the amount of blood will be added to the database according to the blood group. This page also contain table which displays the list of all the donor we can just delete the data of that donor through selecting his name and just delete it.

5. **Receiver Page:**

This a page in which we can add the details of a receiver. In details here we need to take id name and blood group as an input and it is mandatory to give the amount of blood that is going to receive by receiver as later confirmation the amount of blood will be subtracted from the database according to the blood group. This page also contain table which displays the list of all the receivers we can just delete the data of that person through selecting his name and just delete it.

6.View Tables Page:

This is a page which contain the whole data. Once you open this page you will encounter with tabbed pane which contain receiver, donor and storage receiver tab contains the data of all the receiver and while donor contain data of all the donors.
Here we Can See Details Of all the donor and receiver according to their blood groups. And the one last tab is of storage where we can see the availability of all the blood group.

Chapter 4:

Testing Of Software:

Testing of software component and finished products along with quality assurance procedures.

Testing is an important part of software development, and that goes doubly projects as large and complex.

The project that we created must have to test. There will be some errors and its must to resolve that error and develop our project quality assurance by making it error free.

The project testing phase means a group of activities designated for investigating and examining progress of a given project to provide stakeholders with information about actual levels of performance and quality of the project.

The purpose of test phase is to guarantee that system successfully built and tested in the Development Phase meet all requirements and design parameters.

After being tested and accepted, the system moves to the implementation Phase.

First, we tested all the condition in our code and all the cases when user gives invalid input and according to that the software is again transferred to coding phase and again tested. This process is repeated until no bugs are found in our software.

For all of this testing we have carried out white and black box testing.

White Box Texting

The primary objective of white box testing for the Blood Bank Management System is to verify the correctness of the source code, identify potential vulnerabilities, and ensure that the software functions as expected. It also aims to assess code coverage to determine the adequacy of test cases.

Testing Scope

White box testing covered various aspects of the Blood Bank Management System, including:

1. Source code correctness.
2. Code coverage analysis.
3. Error handling and exception testing.
4. Security testing of authentication and authorization mechanisms.

Test Results:

The white box testing of the Blood Bank Management System yielded the following results:

Code Review: The code review identified several coding standards violations and some areas where code optimization was needed. All identified issues were documented and communicated to the development team for resolution.

Static Analysis: The static analysis tools did not find any critical vulnerabilities in the code. However, minor issues such as potential null pointer exceptions were identified and addressed.

Unit Testing: Unit tests were successfully created for all major code modules. The tests provided a high level of code coverage, with over 90% of the code executed during testing.

Security Testing: Authentication and authorization mechanisms were rigorously tested and found to be secure against common security threats, including brute force attacks and SQL injection.

The white box testing conducted on the Blood Bank Management System demonstrated that the software is well-structured and relatively free of critical vulnerabilities. The code review and static analysis helped identify and address coding standards violations and minor issues. The unit tests provided substantial code coverage, and security testing confirmed the system's robustness

Black Box Testing

This report presents the results of the black box testing conducted on the Blood Bank Management System, a critical software application designed to streamline the operations of blood banks and ensure efficient blood donation and distribution processes. The primary objective of this testing was to evaluate the functionality, usability, and security of the system while focusing on its external behavior without knowledge of its internal code structure.

Scope:

The black box testing for the Blood Bank Management System encompassed various test scenarios, including functional, usability, performance, and security testing. The scope of this testing effort aimed to identify potential defects, vulnerabilities, and usability issues within the system.

Testing Results:

1. **Functional Testing Results:** The Blood Bank Management System exhibited robust functionality. All core features, including donor registration, blood inventory management, and request processing, performed as expected. No critical functional defects were identified during testing.

2. **Usability Testing Results:** The system's usability was generally satisfactory, with an intuitive user interface. However, some minor issues related to inconsistent button placement and labeling were identified. These issues are recommended for improvement to enhance the overall user experience.

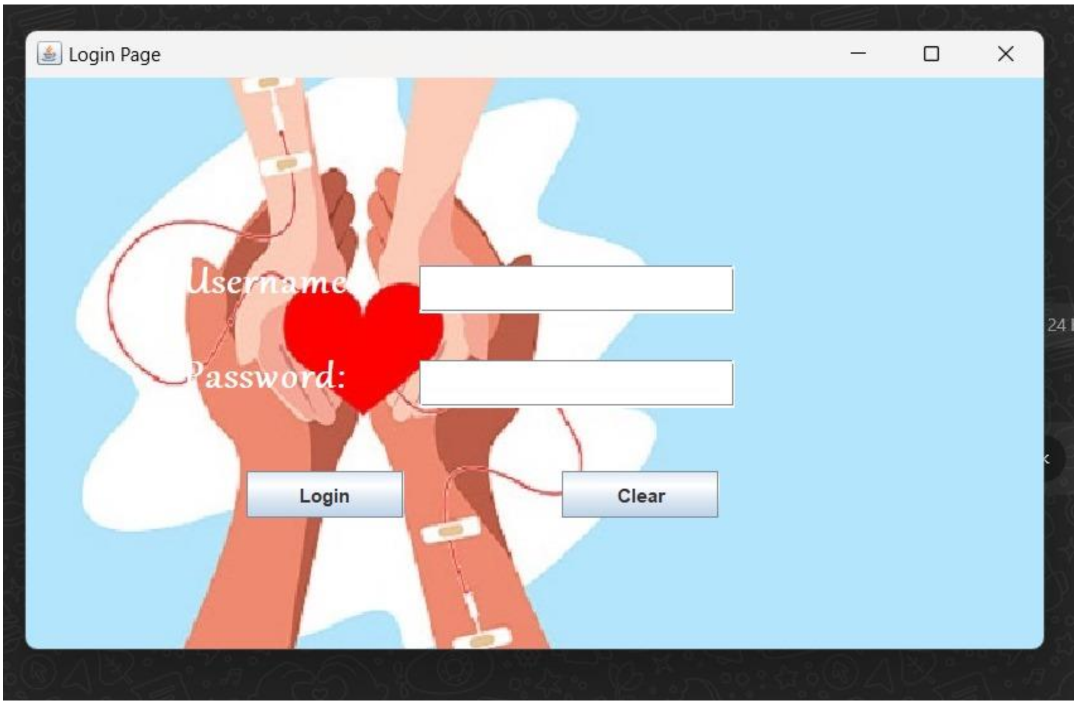
3. **Performance Testing Results:** Performance testing results indicated that the system was responsive under normal and peak load conditions. However, some performance degradation was observed during stress testing, particularly when simulating a high number of concurrent users. This suggests a need for optimization to improve scalability.

4. **Security Testing Results:** The security testing phase revealed some vulnerabilities, including inadequate input validation and authentication flaws. Penetration testing identified potential entry points for malicious actors. These findings highlight the importance of implementing robust security measures to protect sensitive data.

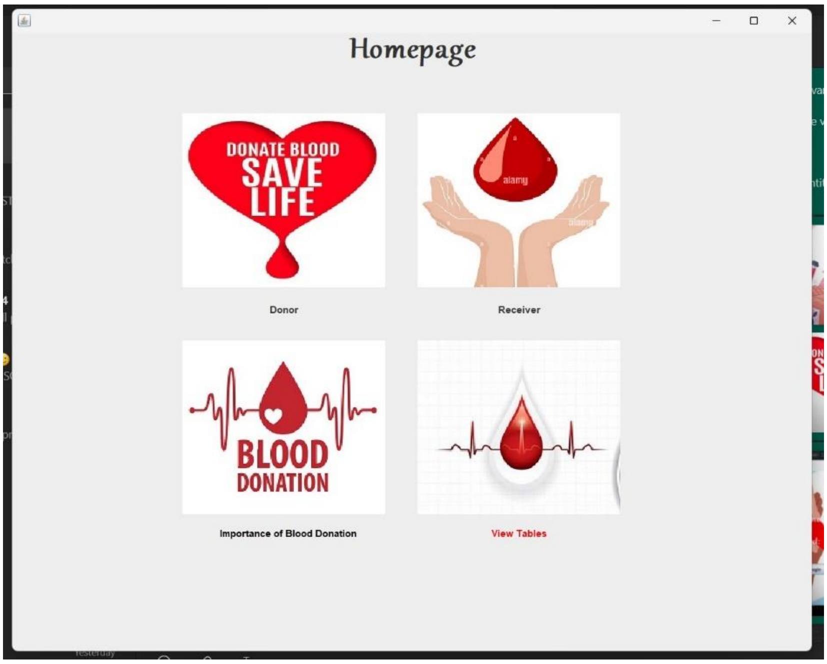
Chapter 5:

Products:

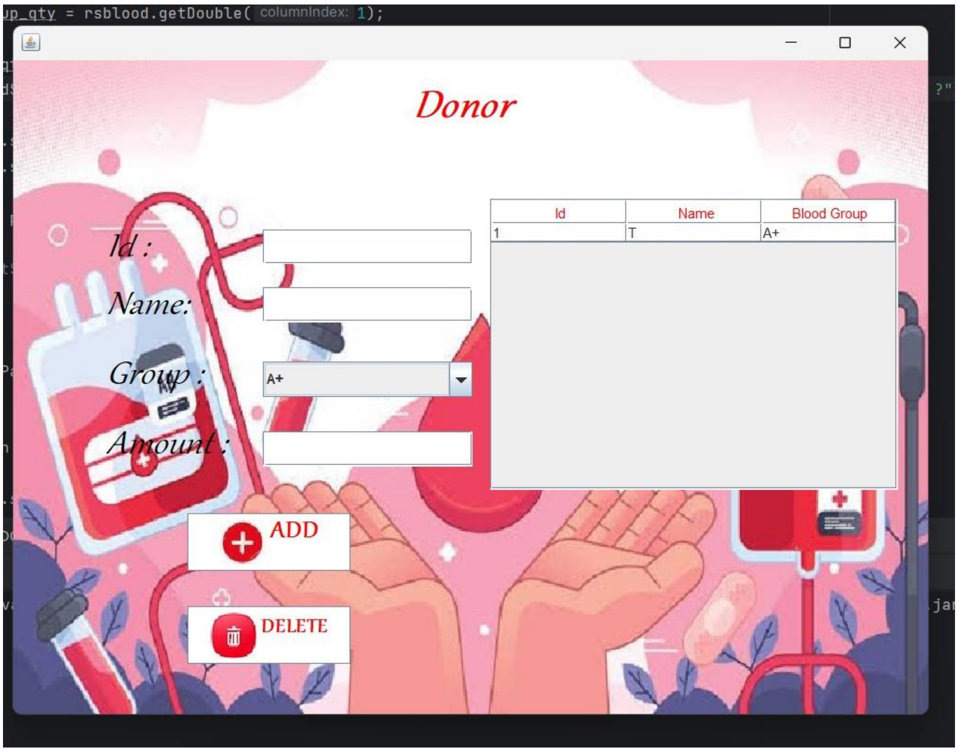
1.Login Page:



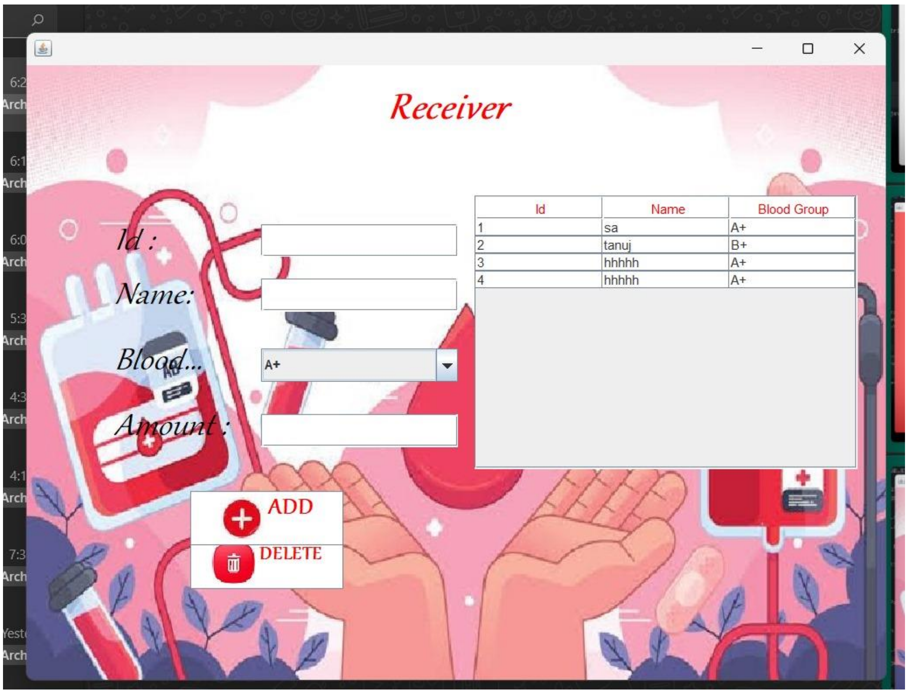
2. Home Page:



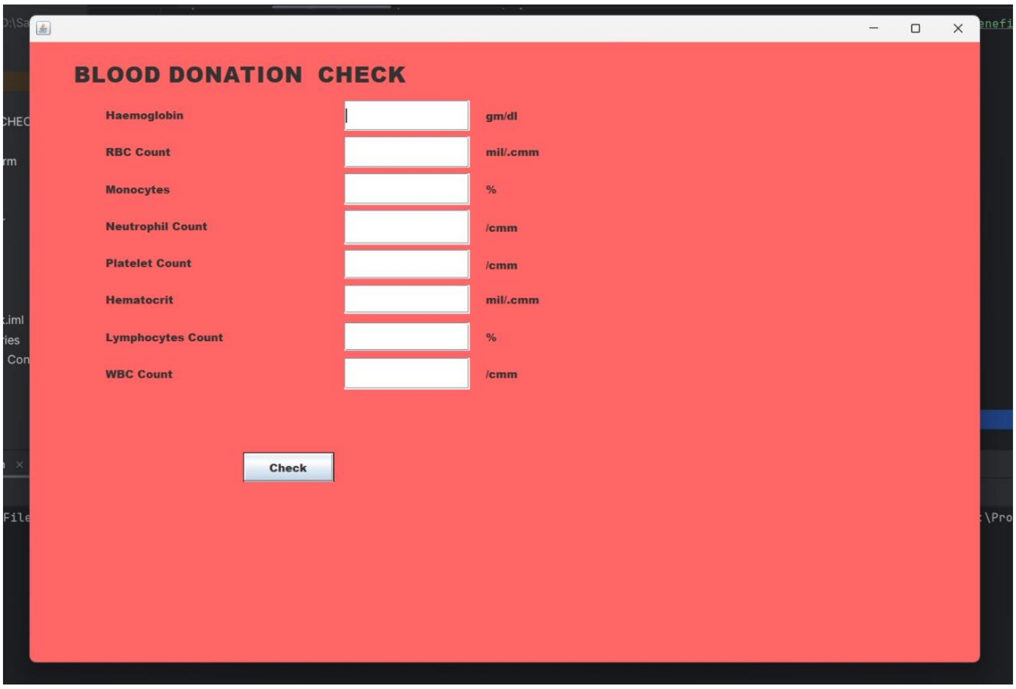
3. **Donar Page:**



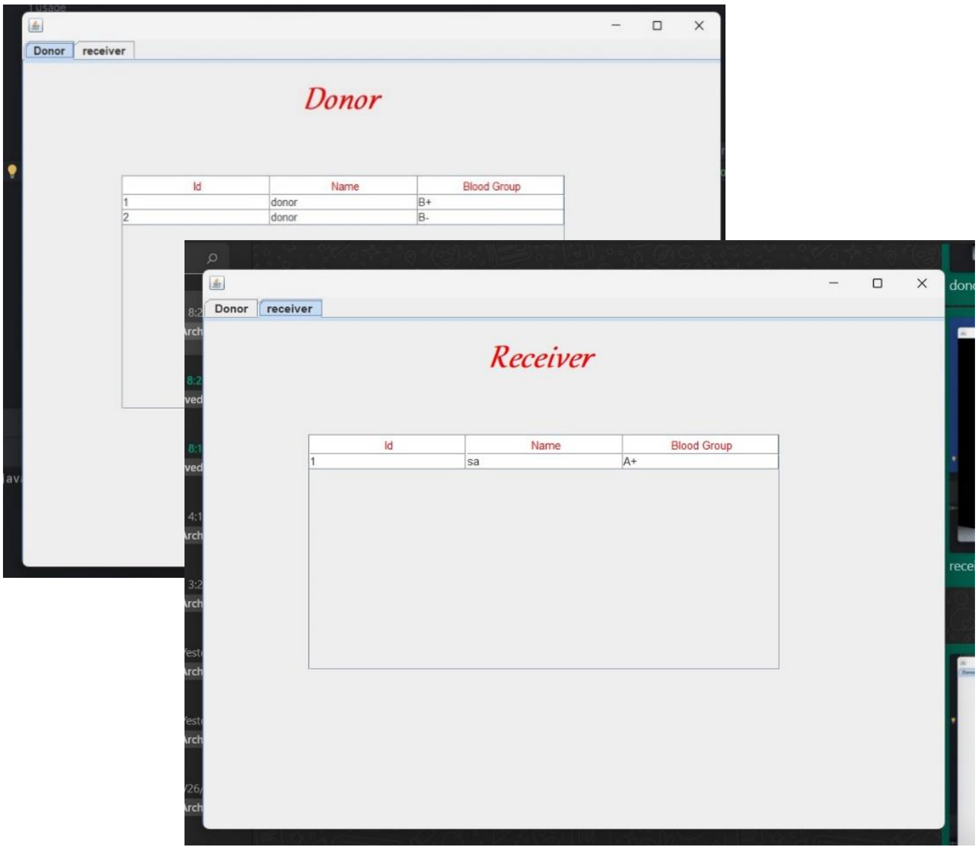
4. **Receiver Page:**



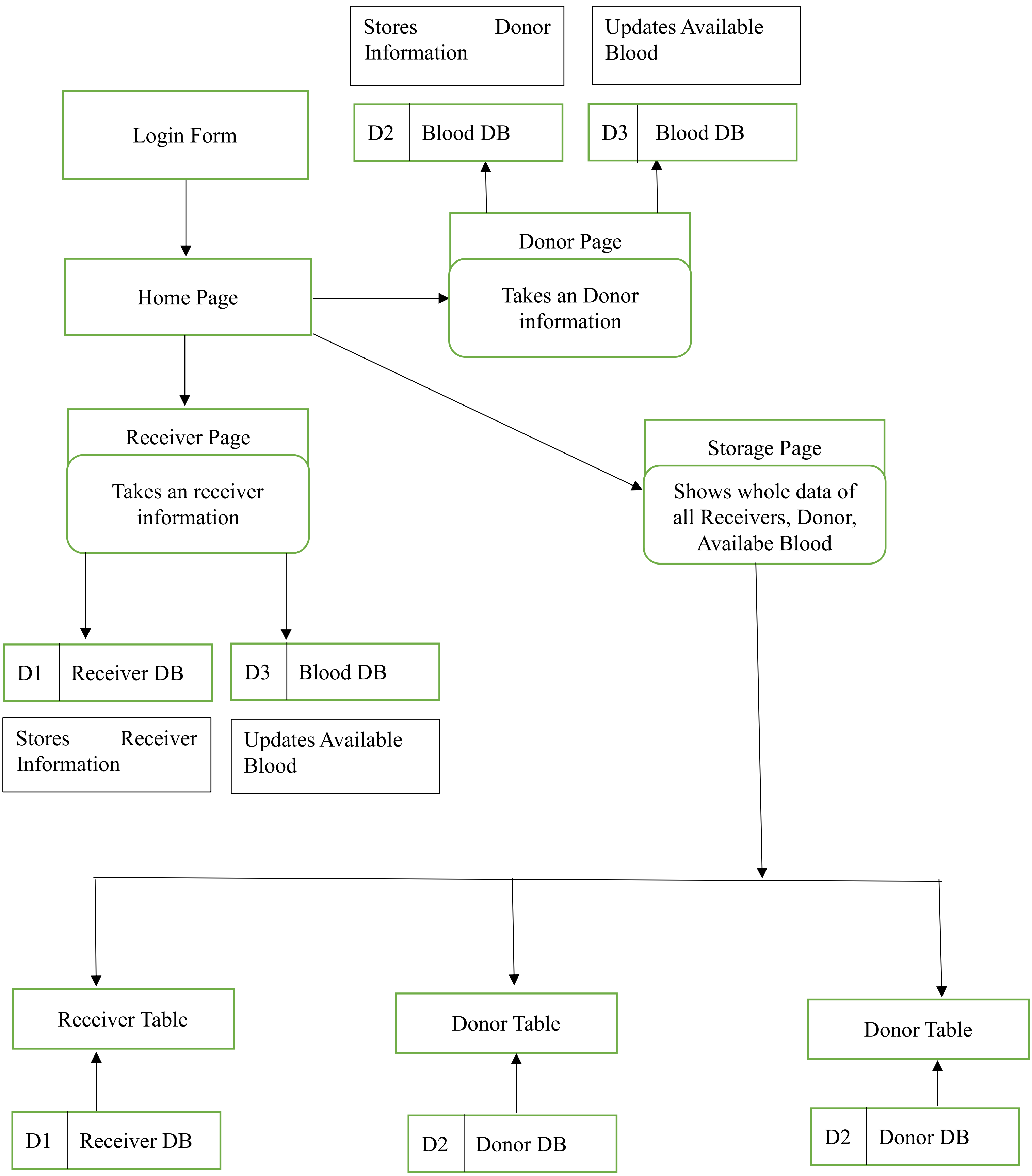
5. **Blood Donation Check Page:**



6.View Tables Page:



Data Flow Diagram



Chapter 6:

Safety Precautions:

In software engineering, software system safety optimizes system safety in the design, development, use, and maintenance of software systems and their integration with safety-critical hardware systems in an operational environment.

A software specification error, design flaw, or the lack of generic safety-critical requirements can contribute to or cause a system failure or erroneous human decision.

To achieve an acceptable level of safety for software used in critical applications, software system safety engineering must be given primary emphasis early in the requirements definition and system conceptual design process.

Safety-critical software must then receive continuous management emphasis and engineering analysis throughout the development and operational lifecycles of the system. Software with safety-critical functionality must be thoroughly verified with objective analysis.

Functional safety is achieved through engineering development to ensure correct execution and behavior of software functions as intended

Safety consistent with mission requirements, is designed into the software in a timely, cost effective manner.

On complex systems involving many interactions safety-critical functionality should be identified and thoroughly analyzed before deriving hazards and design safeguards for mitigations.

Safety-critical functions lists and preliminary hazards lists should be determined proactively and influence the requirements that will be implemented in software.

Chapter 7:

Particulars of practical experiences in Industry/organization if any in My Java-Based Internship Journey

During my internship, I had the privilege of diving headfirst into the world of Java-based software development. This journey not only expanded my technical skills but also broadened my understanding of how Java, as a versatile and powerful programming language, is utilized in real-

From day one, I was immersed in a collaborative environment where learning was paramount. My mentors provided me with comprehensive insights into Java's fundamentals, best practices, and its crucial role in various domains, from web development to mobile app creation.

One of the highlights of my internship was the hands-on experience I gained in building Java applications. I was involved in developing a web-based inventory management system for a small business client. This project not only allowed me to apply my Java skills but also taught me the importance of designing scalable and efficient software. I learned to create responsive user interfaces using JavaFX, manage databases with JDBC, and implement secure authentication mechanisms.

Problem Solving and Debugging

Throughout my internship, I encountered various challenges, from complex logic errors to optimizing code for performance. Each challenge served as an opportunity to sharpen my problem-solving abilities and delve deeper into debugging techniques. I also had the privilege of collaborating with experienced developers, which exposed me to industry-standard debugging practices and the art of writing clean, maintainable code.

Beyond the technical aspects, my internship introduced me to the importance of continuous learning. I explored Java frameworks and libraries like Spring and Hibernate, which opened up new possibilities for enhancing application functionality and scalability. Additionally, I gained exposure to version control systems like Git, which is indispensable in collaborative development.

Chapter 8:

Special experiences encountered during training:

This internship has been an excellent and rewarding experience. I conclude that there has been a lot that I have learnt from my work at Compilers Technologies.

To sum it all up, listed further is the summary of the tasks that I have completed during My Internship at Compilers Technology Learned and brush up our concepts of Java, Swing, SQL, JDBC, Basics of (HTML, CSS, JavaScript), JSP, Servlets

Learned how to work, Learned about connection of java program with database. Learned about how to create a useful software using swing.

Learned about web development servlets, jsp and basic of html, css, js.

Chapter 9:

References/Bibliography:

1.Stack Overflow:

https://stackoverflow.com/questions/28916386/learn-event-handling-in-java

2.GitHub

https://github.com/topics/jframe

3.JavaTpoint

https://www.javatpoint.com/java-jdbc

4.Notes taken from internship

GOVERNMENT POLYTECHNIC, AMRAVATI

(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)

(Format 5)

Daily Diary

For Industrial Training

Name and address of the industry: Compilers Technologies Software and Web Development

Duration: From 08:00 am To 12:00 noon

Name of Industry Personnel: Pravinkumar R. Patil **Designation Of Industry Personnel:** Proprietor

Id Code & Name of the Student: 21IF035 Tanuj Avinash Keshattiwar

Name of Programme: Windows-based BLOOD BANK SYSTEM using java

---------------------------------- ----------------------------------

Signature of Student Signature of Industrial Supervisor

Annexture - 1 (Daily Dairy)

GOVERNMENT POLYTECHNIC, AMRAVATI

(AN AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)

(Format 5)

Week no:1 (From Dt. 19/06/23To Dt. 24/06/23)

Day/Date	**Activities carried out/noted**
19/06/2023	**SWING Classes** class JFrame – An object of this class represents a Frame object (Window object) class JComponent -
20/06/2023	class JLabel – An object of this class can be used to display a static text or an image. class ImageIcon – An object of this class represents an image.
21/06/2023	class JScrollPane – An object of this class represents a scrollable platform. class JSplitPane – An object of this class represents a container which can be divided into two parts.
22/06/2023	class JButton – An object of this class represents a push button which performs sometask when it is clicked. interface ActionListener – This interface must be implemented by the listener who is register with the JButton to handle its click event.

23/06/2023	class ActionEvent – An object of this class contains information about the event source. **Event Delegation Model** – Event Source----Listener ----- Event Handling Method.
24/06/2023	class JTextField – An object of this class represents a single line textbox which can be used to input values. class JPasswordField – An object of this class represents a text box to read password.

Week no: 2 (From Dt. 20/06/23To Dt. 01/07/23)

Day/Date	**Activities carried out/noted**
20/06/2023	class JTextArea – An object of this class represents a multiline textbox. class JToolBar – An object of this class represents a toolbar on which we use options.
27/06/2023	class JMenuBar – An object of this class represents a JMenuBar which can be attached a JFrame class JMenu - An object of this class represents a JMenu which can be added in the JMenuBar
28/06/2023	class JMenuItem - An object of this class represents a JMenuItem which can be added in the JMenu. class JFileChooser - An object of this class can be used to display a dialog box to open or save a file.
29/06/2023	class FileNameExtensionFilter -An object of this class represents a File Filter. class JColorChooser – This class provides us a static method to display color dialog box from which we can select a color.

30/06/2023	class JSlider - An object of this class represents a slider to select a value in the specified range. interface ChangeListener – This interface must be implemented by the Listener object who is register with the JSlider or JProgressBar to handle their value changed events.
01/07/2023	class JProgressBar - An object of this class represents a progress bar to display the progress of some task class JTable - An object of this class represents a tabular grid in which we can display the data in tabular format.

Week no:3 (From Dt. 03/07/23 To Dt. 08/07/23)

Day/Date	**Activities carried out/noted**
03/07/2023	class JTabbedPane – An object of this class represents a tabbed control class JDesktopPane – An object of this class represents a container in which we can add JInternalFrame Objects
04/07/2023	class JInternalFrame – An object of this class represents an internal frame as child frame which can be added in the JDesktopPane class JList – An object of this class represents a List box which contains a list of items and we can select one or more items from the list.
05/07/2023	Interface ListSelectionListener - This interface must be implemented by the listener who is registered with the Jlist to handle its selection changed events. class JChechBox - An object of this class can be used to take boolean input from the user also the checkboxes can be used to select 0 or more options from the list of options.

06/07/2023	class JRadioButton - An object of this class represents a radio button wchich can be used to select single options from the list of options. class ButtonGroup - An object of this class represents a group of related JRadioButton. RadioButton belonging to the same category must have same button group.
07/07/2023	class JComboBox - An object of this class represents a drop down list of items from which we can select one item at a time. class JOptionPane - This class provides different static methods to display input dialog boxes, messages, confirm dialog, and customized option dialog box.
08/07/2023	GridLayout - The GridLayout class is a layout manager that lays out a container's components in a rectangular grid. FlowLayout - A flow layout arranges components in a directional flow, much like lines of text in a paragraph.

Week no:4 (From Dt. 11/07/23 To Dt. 16/07/23)

Day/Date	Activities carried out/noted
11/07/2023	class JApplet - This class is derived from the class Applet and is available in the package Javax.swing.
12/07/2023	Java Database Connectivity (JDBC) - The JDBC technology used to connect java frontend with back end(i.e. Database) and by using this we can connect with database like Oracle,Mysql,MS-Access, etc.
13/07/2023	class - This class provide methods to register a database driver to find the name of the class and package ,etc. class DiverManager - This class provides different static method to register database connector driver and establish connection with the database.
14/07/2023	Interface Connection - An object of this type represents a connection with database. Interface Statement - An object of this type is used to execute SQL query.
15/07/2023	Interface ResultSet - The ResultSet is used to store the result of SELECT query. It acquires the structure as columns and rows return by SELECT query.
16/07/2023	Interface PreparedStatement - The PreparedStatement is used to execute parameterized SQL query.

Week no:5 (From Dt. 17/07/23 To Dt. 22/07/23)

Day/Date	**Activities carried out/noted**
17/07/2023	Servlets- servlet technology is used to generate dynamic web pages. class Generic Servlet – This class must be the direct or indirect base class of all the user defined servlet classes.
18/07/2023	Class Interface ServletRequest – An object of this type is automatically created by the servlet container and passes it as an argument to the service() method. Interface ServletResponse – An object of this type is automatically by the servlet container and passes an argument to the service() method.
19/07/2023	Configuration of Tomcat Basics of HTML and CSS
20/07/2023	Topic Discussion for Internship Project
21/07/2023	Summery for all the topics Submitted
22/07/2023	Topic Finalized –Blood Bank Management System

Week no:6 (From Dt. 24/07/23 To Dt.29/07/23)

Day/Date	Activities carried out/noted
24/07/2023	Main project of Industrial Training …… BLOOD BANK Management System Module1 – Login Page: This is Login Page of our project. The User can login by entering userid and Password. Database- MySQL : it contains various tables like donar, receiver,etc
25/07/2023	Module 2- Home Page: After Login This is the First Page. In this Page we have four buttons and each button having different Functionalities.
26/07/2023	Module3 –Donar Page: This module is used for adding new Donar. In this page There have Multiple Fields to get details of the patient and then we goes to new page for getting donation info of patient.
27/07/2023	Module 4- Receiver: This module is used for adding Receiver details.
28/07/2023	Module 5 –Table Page: This module is used for sh0wing data of all donars and receivers.

29/07/2023	Module 6 –Storage: This module is used for Data of all type of available blood . Module 7 –Importance page: This page use to display the importance of blood.

Annexture - 2

Learning Material collected from Industry.

During the internship, we were introduced to various tools used for Java Development with Exposure to Servlet and HTML and CSS Technology.

HTML

The Hypertext Markup Language or HTML is the standard markup language for documents designed to be displayed in a web browser. It can be assisted by technologies such as Cascading Style Sheets (CSS) and scripting languages such as JavaScript. Web browsers receive HTML documents from a web server or from local storage and render the documents into multimedia web pages. HTML describes the structure of a web page semantically and originally included cues for the appearance of the document.HTML elements are the building blocks of HTML pages. With HTML constructs, images and other objects such as interactive forms may be embedded into the rendered page. HTML is the language for **describing the structure of Web pages**. HTML gives authors the means to: Publish online documents with headings, text, tables, lists, photos, etc. Retrieve online information via hypertext links, at the click of a button. **HTML** is a MUST for students and working professionals to become a great Software Engineer specially when they are working in Web Development Domain.

CSS

Cascading Style Sheets (CSS) is a style sheet language used for describing the presentation of a document written in a markup language such as HTML.CSS is a cornerstone technology of the World Wide Web, alongside HTML and JavaScript. CSS is designed to enable the separation of presentation and content, including layout, colors, and fonts. **C**ascading **S**tyle **S**heets, fondly referred to as CSS, is a simple design language intended to simplify the process of making web pages presentable. CSS handles the look and feel part of a web page. Using CSS, you can control the color of the text, the style of fonts, the spacing between paragraphs, how columns are sized and laid out, what background images or colors are used, layout designs, variations in display for different devices.

Servlet

The servlet is used to create web pages; as an API, which provides interfaces, etc. It is used to extend the capabilities of the server which hosts applications on a request-response programming model. Servlets provide component-based and a platform-independent method to build web-based applications without any performance limitations. Servlets in Java have entire access over Java API's and JDBC to access the enterprise database. We shall see in detail what are these Servlets, why are they used, its advantages and limitations, and how actually servlets work in Java. A servlet is a small Java program that runs within a Web server. Servlets receive and respond to requests from Web clients, usually across HTTP, the Hypertext Transfer Protocol. To implement this interface, you can write a generic servlet that extends java. servlet. A Jakarta Servlet is a Java software component that extends the capabilities of a server. Although servlets can respond to many types of requests, they most commonly implement web containers for hosting web applications on web servers and thus qualify as a server-side servlet web API

JSP

JSP stands for Java Server Pages is a technology for building web applications that support dynamic content and acts as a Java servlet technology. You can consider it a different option to a servlet, and it has a lot more capabilities than a servlet. JSP is used explicitly for creating dynamic web applications. Pages created using JSP are simpler to manage as compared to a Servlet. JSP pages are the reciprocal of Servlets as a servlet incorporates HTML code within Java code, whereas JSP includes Java code within HTML with the help of JSP tags. JSP can do all the things that a Servlet can do. Jakarta Server Pages is a collection of technologies that helps software developers create dynamically generated web pages based on HTML, XML, SOAP, or other document types. Released in 1999 by Sun Microsystems, JSP is similar to PHP and ASP, but uses the Java programming language.

Oracle Database

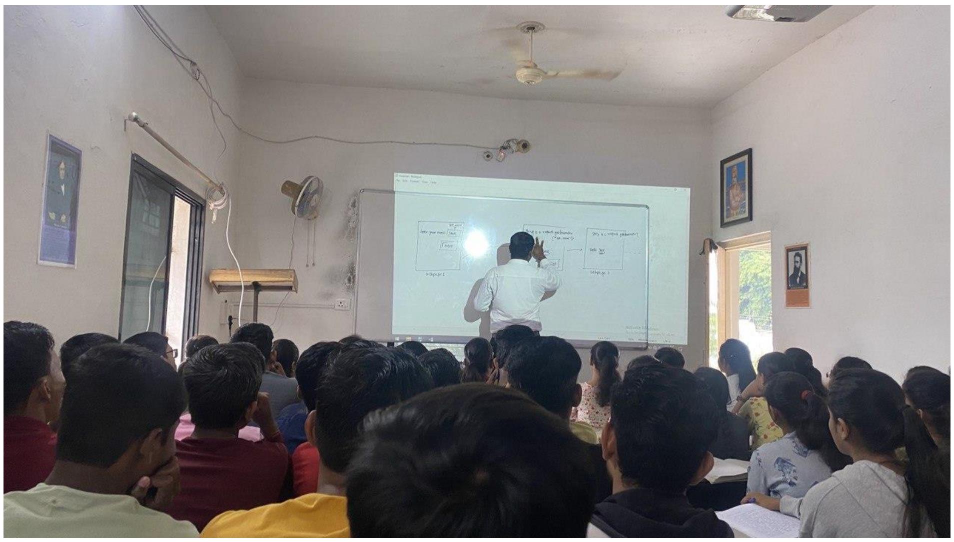
Oracle Database Architecture. An Oracle database is a collection of data treated as a unit. The purpose of a database is to store and retrieve related information. A database server is the key to solving the problems of information management. Oracle Database allows you to quickly and safely store and retrieve data. Here are the integration benefits of the Oracle Database Oracle Database is crossplatform. It can run on various hardware across operating systems including Windows Server, Unix, and various distributions of GNU/Linux. Oracle Database has its networking stack that allows application from a different platform to communicate with the Oracle Database smoothly. For example, applications running on Windows can connect to the Oracle Database running on Unix..

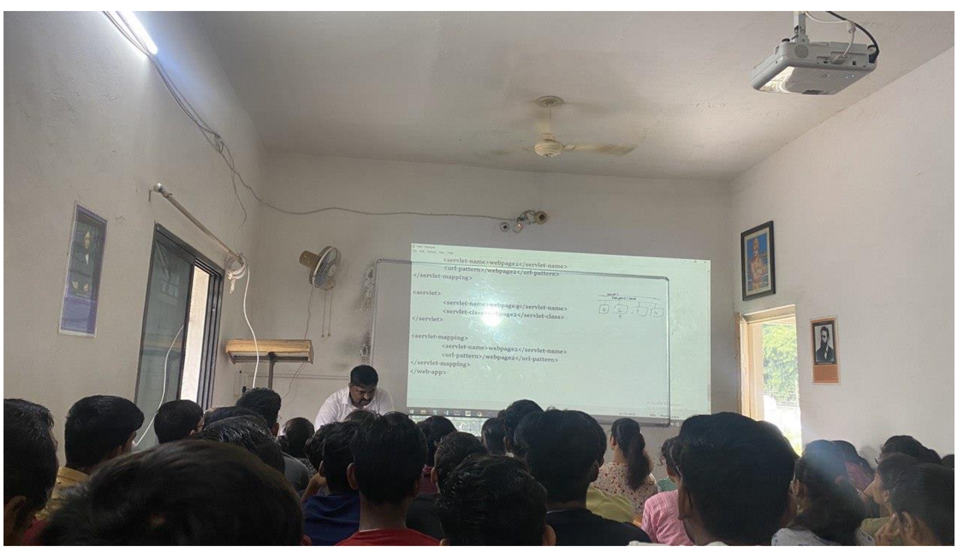
Apache Tomcat

Apache Tomcat is different from Apache you simply need to understand this simple concept: Apache Tomcat is a Java servlet container which works as a webpage server. Remember, a Java servlet is a particular type of Java program that can extend the capabilities of a specific server. Java servlets can respond to any type of request but Java servlets are generally used to enable applications which are hosted on a web server. Apache Tomcat is a popular open-source web server and Servlet container for Java code. As the reference implementation of Java Servlet and Java Server Pages (JSP), Tomcat was started at Sun Microsystems, which later donated the code base to the Apache Software Foundation. It is an open-source Java servlet container that implements many Java Enterprise Specs such as the Websites API, Java-Server Pages and last but not least, the Java Servlet. The complete name of Tomcat is "Apache Tomcat" it was developed in an open, participatory environment and released in 1998 for the very first time. It began as the reference implementation for the very first Java-Server Pages and the Java Servlet API.

Annexture – 3

Photographs





FUTURE SCOPE

The Library Management System is being developed as an application. Normally all users will surely get benefits over it. But in future by considering the mobile users in mind one mobile version (Mobile app) would be build. This will be very helpful for those users who will not aware of handling or faces difficulties while working of it.

CONCLUSION

Overall, This Internship Was A Useful Experience. I Have Gained New Knowledge, Skills and Meet Many New People. Two Main Things That I've learned. The importance Of Our Time Management Skills and Self-motivation. Achieved Several of My Learning Goals, However For some The Conditions Did Not Permit. I Got Insight into Professional Practice. The Internship Was Also Good to Find Out What My Strengths and Weaknesses Are. This helped me to Define what skills and knowledge. I must improve in the coming time.

It Would Be Better That the Knowledge Level of The Language Is Sufficient to Contribute Fully to Projects. While Working with A Company You Must Work with Another People, And It Is a Team Work So Get Experience Of Working With Them Which Will Be Helpful To Me In Further Times Whenever I'll Work With The Team. At Last, This Internship Has Given Me New Insights and Motivation About Working in The Industry And It Will Be More Helpful To Me When I'll Go For A Job After My Graduation.